

Syllabus

II – Semester – [Chemistry]

Examination Scheme for

CA – Continuous Assessment

EoSE - End of Semester Examination

For Regular Students –

Type of Examination	Course Code / Nomenclature	Duration of Examination		Maximum Marks		Minimum Marks	
Theory	CHM-52T-103 – Chemical Bonding, Reactions Mechanism, Aromatic & Aliphatic hydrocarbons, Alkyl & Aryl Halides, States of Matter.	CA	1 Hr.	CA	20	CA	8
		EoSE	3 Hrs.	EoSE	80	EoSE	32
Practical	CHM-52P-104 – Practical II	CA	1 Hr.	CA	10	CA	4
		EoSE	4 Hrs.	EoSE	40	EoSE	16

The question paper in EoSE (End of Semester Examination) will consist of two parts A & B.

PART – A: 20 Marks

Part A will be compulsory, consisting of 10 very short answer-type questions (with a limit of 20 words) covering the entire syllabus, carrying two marks for each.

PART – B: 60 Marks

Part B of the question paper will have four questions with internal choice comprising of question number 2-5 which will be divided into four units. There will be one question with internal choice from each unit. Each question will carry 15 marks.

For Non-Collegiate Students –

Type of Examination	Course Code and Nomenclature	Duration of Examination	Maximum Marks	Minimum Marks
Theory	CHM-52T-103 – Chemical Bonding, Reactions Mechanism, Aromatic & Aliphatic hydrocarbons, Alkyl & Aryl Halides, States of Matter.	3 Hrs.	100	40
Practical	CHM-52P-104 Practical II	4 Hrs.	50	20

The question paper in EoSE (End of Semester Examination) will consist of two parts A & B

PART – A: 20 Marks

Part A will be compulsory, consisting of 10 very short answer-type questions (with a limit of 20 words) covering the entire syllabus, carrying two marks for each.

PART – B: 80 Marks

Part B of the question paper will have four questions with internal choice comprising of question number 2-5 which will be divided into four units. There will be one question with internal choice from each unit. Each question will carry 20 marks.

Syllabus

II – Semester – [Chemistry]

[UG0802/03]-[CHM-52T-103]- Chemical Bonding, Reactions Mechanism, Aromatic & Aliphatic hydrocarbons, Alkyl & Aryl Halides, States of Matter.

[UG0802/03]-[CHM-52P-104]- PRACTICAL-II

II – Semester – [Chemistry]

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits	
II	CHM-52T-103	Chemical Bonding, Reactions Mechanism, Aromatic & Aliphatic hydrocarbons, Alkyl & Aryl Halides, States of Matter.			5	4	
II	CHM-52P-104	PRACTICAL-II			5	2	
Level of Course	Type of the Course	Credit Distribution			Offered to NC Students	Course Delivery Method	
		Theory	Practical	Total			
5	Major	4	2	6	Yes	Through Lecture, Sixty (60) Lectures	Class room Teaching/Power-Point (PPT)
List of Programme Codes in which offered as Minor Discipline		NA					
Prerequisites/Eligibility		Every student automatically promoted from I to II semester after the I-Semester EoSE.					
Course Objectives:		The main objective of this course is to provide students with a theoretical and conceptual understanding of chemical & metallic bonding of compounds. Structure and reactivity of aromatic & aliphatic hydrocarbons (alkanes, alkenes, cycloalkenes, dienes and alkynes) including alkyl & aryl halides chemistry are also included to enrich knowledge in the fields. Further, our aim is to explain the various states of matter, viz. solid, liquid and gases in this curriculum. The laboratory course is designed to provide practical skill and knowledge to synthesize various organic derivatives including chromatographic techniques. Moreover, qualitative analytical techniques like identification of cations and anions in inorganic mixtures with special anions combinations along with quantitative analysis to determine the order, molecularity and kinetic parameters of reactions are also incorporated to enrich knowledge in the practical field.					

Detailed Syllabus

II – Semester – [Chemistry]

CHM-52T-103-Chemical Bonding, Reactions Mechanism, Aromatic & Aliphatic hydrocarbons, Alkyl & Aryl Halides, States of Matter.

Unit-I

Ionic Solids: Ionic structures, Radius ratio effect and coordination number, Limitations of radius ratio rule, Lattice defects, Semiconductors, Lattice energy and Born-Haber cycle, Solvation energy and solubility of ionic solids, Polarizing power and polarisability of ions, Fajan's rule.

Covalent Bond: Valence bond theory and its limitations, Directional characteristics and shapes of simple inorganic molecules and ions. Valence shell electron pair repulsion (VSEPR) theory to NH_3 , H_3O^+ , SF_4 , ClF_3 , ICl_2 , H_2O .

Molecular Orbital Theory: Homonuclear and heteronuclear (CO and NO) diatomic molecules. Multicenter bonding in electron deficient molecules, Bond strength and bond energy, Percentage ionic character from dipole moment and electronegativity difference.

Metallic bond: Free electron, Valence bond and band theories.

Weak Interactions: Hydrogen bonding, Van der Waals forces.

15 Lecture

Unit-II

Mechanism of Organic Reactions:

Reaction intermediates: Types, shapes and relative stability; Carbocations, Carbanions, Free radicals, Carbenes, Nitrenes and Benzyne (Arenes). Types of organic reactions and their mechanism: Addition, Elimination and Substitution reactions. Markovnikov rule, Anti-Markovnikov rule, Saytzeff's rule and Hofmann elimination. Energy considerations. Methods of determination of reaction mechanism (product analysis, intermediates, isotope effects, kinetic and stereochemical studies).

Aliphatic Hydrocarbon

Alkanes and Cycloalkanes: Free radical halogenation of Alkanes: mechanism, orientation, reactivity and selectivity. Cycloalkanes - nomenclature, methods of formation, chemical reactions. Baeyer's strain theory and its limitations. Theory of strain-less rings.

Alkenes, Cycloalkenes, Dienes and Alkynes: Relative stabilities of alkenes. Chemical reactions of alkenes - hydroboration-oxidation, oxymercuration-reduction, epoxidation, ozonolysis and oxidation with $KMnO_4$, Polymerization of alkenes, Substitution at the allylic and vinylic positions of alkenes.

Classification and nomenclature of isolated, conjugated and cumulated dienes. Structure of allenes and butadiene. Methods of formation, properties and chemical reactions - 1,2- and 1,4-additions, Diels-Alder reaction and polymerization reactions.

Structure and bonding of alkynes, Synthetic methods, Chemical reactions - acidity of alkynes, Mechanism of electrophilic and nucleophilic addition reactions: hydroboration-oxidation, metal-ammonia reduction, oxidation and polymerization.

15 Lecture

Unit-III

Arenes and Aromaticity: Nomenclature of benzene derivatives: The aryl group, aromatic nucleus and side chain, Structure of benzene: molecular formula and kekule structure, Stability and carbon-carbon bond lengths of benzene, Resonance structure, MO diagram.

Aromaticity: Huckel's rule, aromatic ions-three to eight membered.

Aromatic electrophilic substitution: General pattern of the mechanism, role of sigma and pi complexes, Mechanism of nitration, halogenation, sulphonation, mercuration, chloromethylation and Friedel crafts reactions, Energy profile diagrams, Activating and deactivating substituents, Directive influence, orientation and ortho/para ratio. Side chain reactions of benzene derivatives, Birch reduction.

Alkyl Halides: Synthetic methods of alkyl halides, Chemical reactions, Mechanisms of nucleophilic substitution reactions of alkyl halides S_N2 and S_N1 reactions with energy profile diagrams.

Polyhalogen compounds: Chloroform, Carbon tetrachloride

Aryl Halides: Methods of formation of aryl halides, Nuclear and side chain reactions, The addition-elimination and the elimination-addition mechanisms of nucleophilic aromatic substitution reactions, Relative reactivities of alkyl, allyl, vinyl and aryl halides.

Unit- IV

States of matter

Gaseous States: Postulates of kinetic theory of gases, Deviation from ideal behavior, Van der Waals equation of state. **Critical Phenomenon:** PV isotherms of real gases, Continuity of states, the isotherms of Van der Waals equation, Relationship between critical constants and van der Waals constants, The law of corresponding states, Reduced equation of state.

Molecular Velocities: Root mean square, average and most probable velocities. Qualitative discussion of the Maxwell's distribution of molecular velocities, Collision number, Mean free path and collision diameter. Liquification of gases (based on Joule-Thomson effect.)

Liquid State: Intermolecular forces, Structure of liquids (a qualitative description). Structural differences between solids, liquids and gases. Liquid crystals: Difference between liquid crystal, solid and liquid. Classification, structure of smectic, nematic and cholesteric phases. Thermography and seven segment cells.

Solid State: Definition of space lattice, Unit cell.

Laws of crystallography- (i) Law of constancy of interfacial angles (ii) Law of rationality of indices (iii) Law of symmetry. Symmetry elements in crystals.

Basic concept of X-ray diffraction by crystals. Derivation of Bragg's equation. Determination of crystal structure of $NaCl$ and $CsCl$ (Laue's method and powder method.), Band theory of solids, Defects in solids.

15 Lectures

Suggested Books and References:

1. Concise Inorganic Chemistry by J.D. Lee, Wiley.
2. Inorganic Chemistry: Principles of Structure and Reactivity by Ellen A. Keiter, James E. Huheey and Richard L. Keiter, Pearson.
3. Organic Chemistry by S. S. Gupta, Oxford University Press.
4. Organic Reaction Mechanisms by V. K. Ahluwalia and Rakesh Kumar Parashar, Narosa Publishing House, New Delhi.
5. Organic Chemistry – Reactions and Reagents: Covering Complete Theoretical Organic Chemistry by O. P Agarwal, Goel Publishing House, Meerut.
6. Organic Chemistry by R. T. Morrison and R. N. Boyd, Prentice Hall.
7. Organic Chemistry by I. L. Finar (Vol. I & II), ELBS.
8. Advanced Organic Chemistry by A. Bahl and B. S. Bahl, S. Chand.
9. Modera Organic Chemistry by M.K. Jain and S. C. Sharma, Vishal Publishing Co.
10. March's Advanced Organic Chemistry: Reactions,
11. Mechanisms and Structure by J. March and M. B. Smith, Wiley. Mechanism in Organic Chemistry by Peter Sykes, Pearson Education.
12. Principles of Physical Chemistry by B. R. Puri, L. R. Sharma and M. S. Pathania, Vishal Publishing Co.
13. Advanced Physical Chemistry by Gurdeep Raj, Goel Publishing House.
14. A Textbook of Physical Chemistry by K. L. Kapoor (Volume 5), Macmillan India Ltd.
15. Chemical Kinetics and Reaction Dynamics by Santosh K. Upadhyay, Springer (Anamaya Publishers, New Delhi, India).

Suggested E-resources:

All the above suggested books are **available as e- books**.

Online Lecture Notes and Course Materials:

All prescribed courses are available in the form of **e-books**, Adobe Acrobat documents (PDF), web pages

Syllabus

CHM-52P-104: Practical II

4 Hrs./week

Inorganic Chemistry

10 marks

1. Qualitative Analysis

Separation and identification of six radicals (three cations and three anions) in the given inorganic mixture including special combinations of anions.

Organic Chemistry

10 marks

2. (a) Synthesis of semi carbazones of any one of the following compounds: acetone, ethyl methyl ketone, cyclohexanone, benzaldehyde.
- (b) Synthesis of 2,4-dinitrophenylhydrazones of any aldehyde /ketone.
- (c) Synthesis of Coumarin through salicylaldehyde.
- (d) Synthesis of 3,5- dinitro benzoic acid using benzoic acid and nitrating reagent

OR

- (a) Chromatography separation of the active ingredients of plants (spinach etc.), flowers and juices by Paper/Thin layer chromatography (TLC).
- (b) Separation and identification of the monosaccharides in the given mixture by Paper/Thin layer chromatography (TLC) and reporting the R_f values.
- (c) Separation of a mixture of two amino acids by ascending and horizontal paper chromatography.

Physical Chemistry

10 marks

3. Chemical Kinetics:

- (a) To determine the specific reaction rate of the hydrolysis of methyl acetate/ethyl acetate catalyzed by hydrogen ions at room temperature.
- (b) To study the effect of acid strength on the hydrolysis of an ester.
- (c) To compare the strengths of HCl and H_2SO_4 by studying the kinetics of hydrolysis of ethyl acetate.
- (d) To study the oxidation of iodide ions by H_2O_2 as an iodine clock reaction.
- (e) To study the decomposition of H_2O_2 catalyzed by iodide ions.
- (f) To study the kinetics of reaction between acetone and iodine in presence of an acids.

4. Viva voce

5 marks

5. Practical Record

5 marks

Suggested Books and References:

1. Vogel's Qualitative Inorganic Analysis, A. I. Vogel Prentice Hall.
2. Vogel's Quantitative Inorganic Analysis Including Elementary Instrumental Analysis, ELBS.
3. Vogel's Textbook of Quantitative Chemical Analysis, A. I. Vogel, Pearson Education Ltd.
4. Advanced Practical Organic Chemistry by N. K. Vishnoi, Vikas Publishing House Pvt Ltd.
5. Comprehensive Practical Organic Chemistry: Preparation and Quantitative Analysis, V. K Ahluwalia. Universities Press, Hyderabad.
6. Laboratory Techniques in Organic Chemistry by V. K Ahluwalia, I K International, N
7. Advanced Practical Organic Chemistry J. B Yadav, Goel Publishing House.
8. Practical Physical Chemistry, by B. D Khosla, S. Chand & Company.
9. Advanced Practical Organic Chemistry by Amit Arora, Discovery Publishing House, New Delhi.

Suggested E-resources:

All the above suggested books are available as e- books.

Online Lecture Notes and Course Materials:

All prescribed courses are available in digital form in the form of e-books, Adobe Acrobat documents (PDF), web pages etc.

Course Learning Outcomes:

Once completed course, students will acquire a clear understanding of chemical bonding, organic reactions mechanisms for various fundamental reactions of aliphatic & aromatic hydrocarbons, alkyl & aryl halides along with states of matters.

They will acquire practical skill and knowledge in qualitative analytical techniques, viz. identification of cations and anions in inorganic mixtures with special anions combinations along with synthetic route of various organic derivatives including chromatographic techniques. Students can achieve practical knowledge about order, molecularity and kinetic parameters of reactions.